



Quantifying Biological Responses

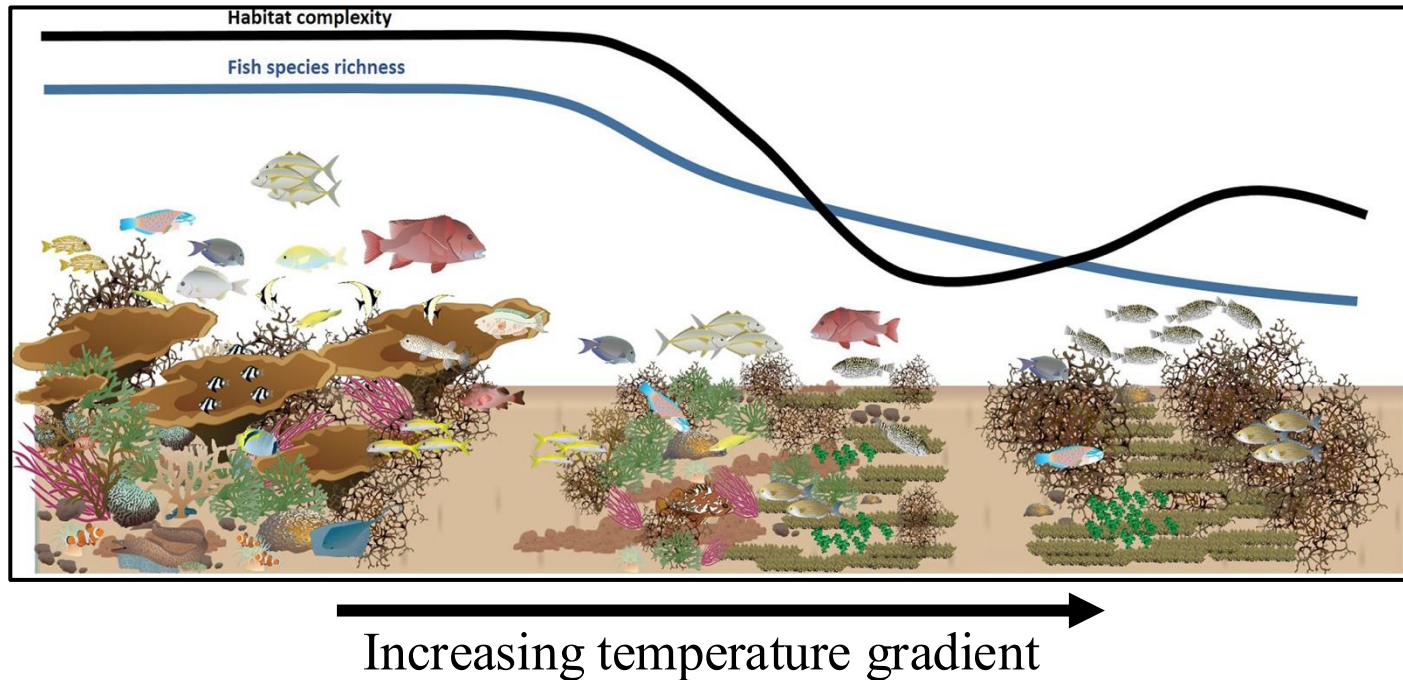
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QUANTIFYING BIOLOGICAL COMMUNITY CHANGES

EFFECTS OF OCEAN WARMING AT COMMUNITY LEVEL



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Ocean warming increases threat of invasive species
in a marine fouling community

CASCADE J. B. SORTE,^{1,3} SUSAN L. WILLIAMS,¹ AND ROBYN A. ZEREBECKI²

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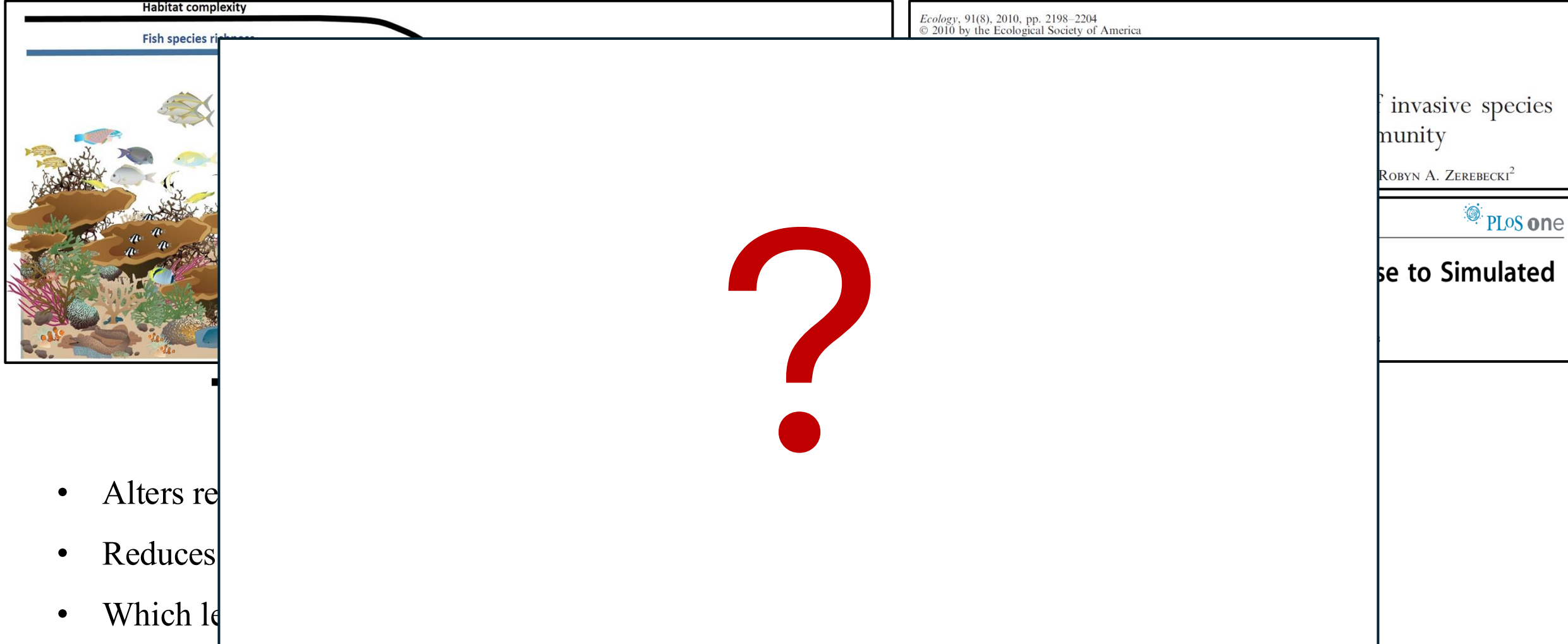
PLoS one

Turning on the Heat: Ecological Response to Simulated
Warming in the Sea

Dan A. Smale^{1*}, Thomas Wernberg^{1,2}, Lloyd S. Peck³, David K. A. Barnes³

- Alters recruitment (addition of new individuals to the community)
- Reduces species diversity and abundance
- Which leads to changes in overall community structure
- Simplified communities, dominated by species with broad thermal tolerance ranges (i.e. invasive species)
- Which could facilitate competition, where some species are outcompeted by others

EFFECTS OF OCEAN WARMING AT COMMUNITY LEVEL

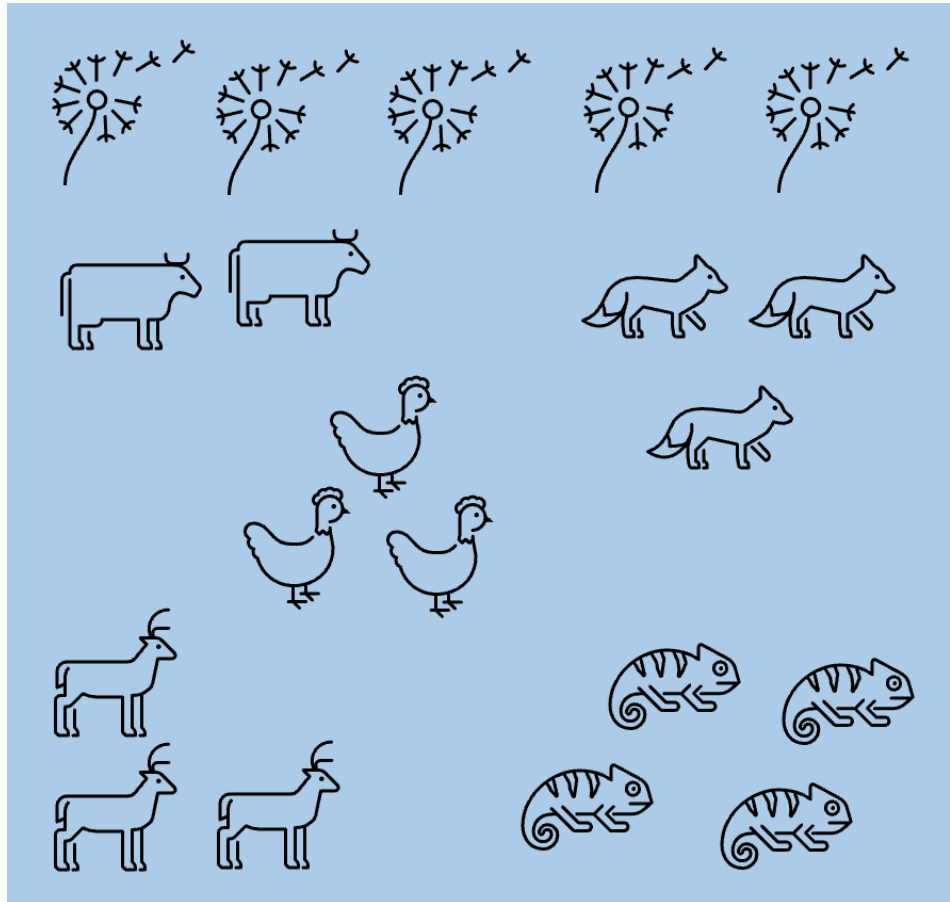


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QUANTIFYING BIOLOGICAL COMMUNITY CHANGES

- Biological Community – interacting group of various species in an area
- Community structure / composition ~ Species Richness + Species Diversity

Community 1



Community 2



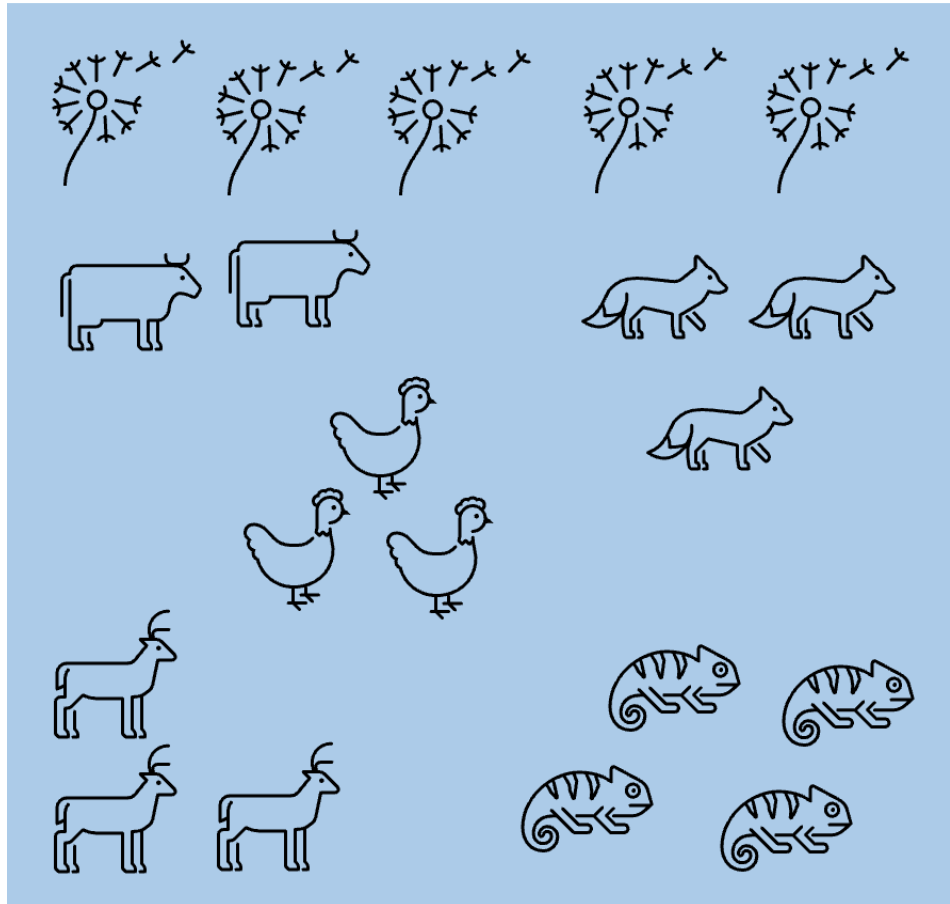
Species Richness

- The number of different species in a particular community
- Does not take into account abundance (number of individuals) of each species

Species	Community 1 abundance	Community 2 abundance
Woodrush	2	7
Holly	8	6
Bramble	1	0
Yorkshire Fog	1	10
Sedge	3	386
Total	15	409

Species Richness

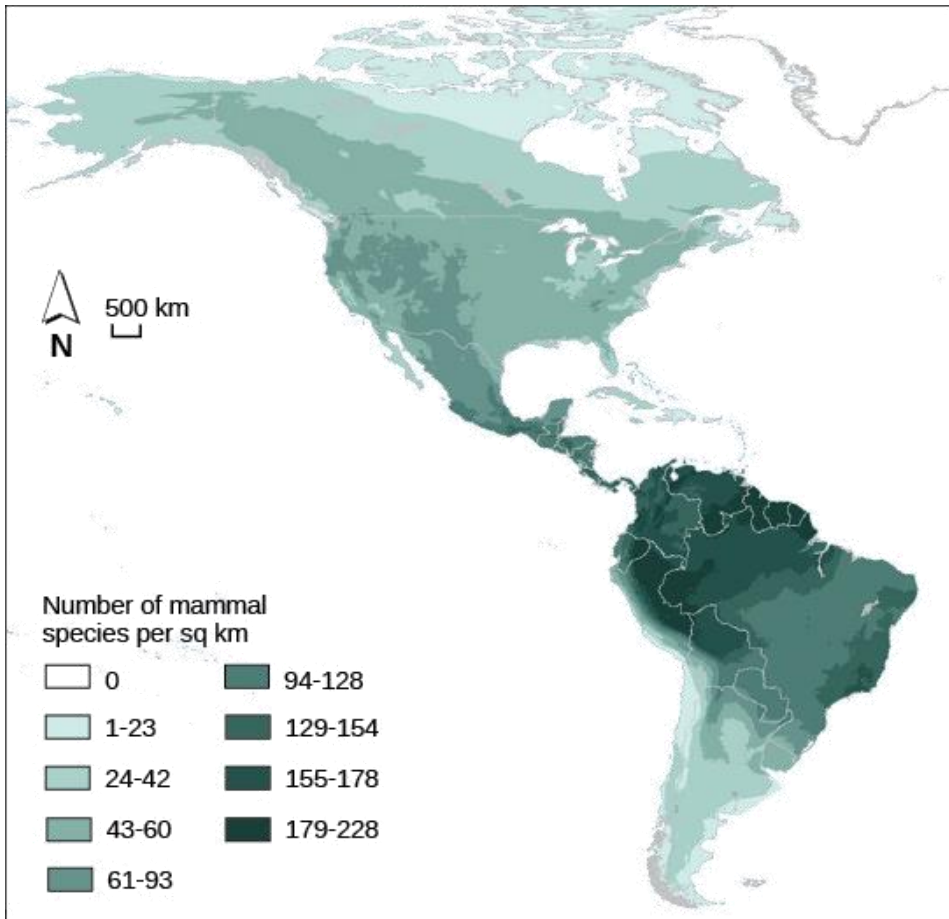
Community 1: Species Richness = 6



Community 2: Species Richness = 5



Species Richness



- Highest species richness tend to be near the equator
 - Lots of solar energy (supporting high primary productivity), warm temperatures, high precipitation, and small seasonal variability
- Lowest species richness lie near the poles
 - Less solar energy and colder, drier

Species Diversity

- Measure of community complexity
- Provides information about rarity or commonness of species
- A function of species richness and species evenness (relative abundances of species)
- As species richness and evenness increase, diversity increases
- Common diversity indices ~ Simpson's Diversity Index and Shannon-Wiener Diversity Index

Species Diversity

Simpson's Diversity Index (D or λ)

- Measures the probability that two randomly chosen individuals belong to the same species
- Measures dominance and focuses on common species
- Values range from **0 to 1**
 - **Closer to 1:** High biodiversity ~ many different species evenly spread out
 - **Closer to 0:** Low biodiversity ~ a single species dominates the area

$$D = 1 - \frac{\sum n(n-1)}{N(N-1)}$$

- **n:** Number of individuals in a single species
- **N:** Total number of individuals in the entire community or sample

Species Diversity

Simpson's Diversity Index (D or λ)

$$D = 1 - \frac{\sum n(n-1)}{N(N-1)}$$

Species	Location A quadrat			Location B quadrat		
	<i>n</i> (number of individuals)	<i>n</i> -1	<i>n</i> (<i>n</i> -1)	<i>n</i>	<i>n</i> -1	<i>n</i> (<i>n</i> -1)
Sea couch grass	21	20	420	18	17	306
Sea twitch grass	1	0	0	11	10	110
Sea rocket	8	7	56	8	7	56
Prickly saltwort	1	0	0	3	2	6
Lyme grass	12	11	132	7	6	42
<i>N</i> (total number of individuals)	43			47		
		Σ	608		Σ	520

For Location A:

$$D = 1 - \frac{608}{43 \times 42}$$

$$D = 1 - \frac{608}{1806}$$

$$D = 1 - 0.337$$

$$D = 0.663$$

For Location B:

$$D = 1 - \frac{520}{47 \times 46}$$

$$D = 1 - \frac{520}{2162}$$

$$D = 1 - 0.241$$

$$D = 0.759$$

Species Diversity

Shannon-Wiener Diversity Index (H)

- Measures the uncertainty in predicting the species of a random individual
- Measures richness and rare species and treats all groups with a broader balance
- Values usually range from **1.5 to 3.5**
 - Higher values ~ greater diversity and a more even distribution of species
 - Lower values ~ lower diversity or a community heavily dominated by a single species
 - Value of zero occurs only if there is exactly one species present in the sample

$$H' = - \sum_{i=1}^S p_i \ln(p_i)$$

Species Diversity

Shannon-Wiener Diversity Index (H)

$$H' = - \sum_{i=1}^S p_i \ln(p_i)$$

- $H' \sim$ **Shannon-Wiener Index value**
- $S \sim$ **total number of species** (richness)
- $p_i \sim$ **proportion of individuals** belonging to the i^{th} species ($p_i = \frac{n_i}{N}$), where n_i is the count of species i and N is the total number of individuals across all species
- **ln** = Natural logarithm (base e), though base 2 or base 10 are sometimes used in specific contexts

Species Diversity

Shannon-Wiener Diversity Index (H)

$$H' = - \sum_{i=1}^S p_i \ln(p_i)$$

A community sample containing
3 different species:

- Species A: 50 individuals
- Species B: 30 individuals
- Species C: 20 individuals

$$N = 50 + 30 + 20 = 100$$

1. Proportions (p_i)

- $p_A = \frac{50}{100} = 0.50$
- $p_B = \frac{30}{100} = 0.30$
- $p_C = \frac{20}{100} = 0.20$

2. $p_i \ln(p_i)$ for Each Species

- **Species A:** $0.50 \times \ln(0.50) \approx -0.3466$
- **Species B:** $0.30 \times \ln(0.30) \approx -0.3612$
- **Species C:** $0.20 \times \ln(0.20) \approx -0.3219$

3. Sum the Values and Invert the Sign

$$\sum = (-0.3466) + (-0.3612) + (-0.3219) = -1.0297$$

$$H' = -(-1.0297) = \mathbf{1.0297}$$

Species Diversity

Shannon-Wiener Diversity Index (H)

$$H' = - \sum_{i=1}^S p_i \ln(p_i)$$

$$H' = -(-1.0297) = 1.0297$$

- **High Diversity ($H > 3.0$):** Indicates an ecosystem with high species richness and even distribution. If you pick an individual at random, it is difficult to predict which species it will be.
- **Low Diversity ($H < 1.5$):** Indicates an ecosystem with few species or one heavily dominated by a single species.
- **Zero Diversity ($H = 0$):** Occurs only when a sample contains exactly **one** species ($p_1 = 1.0$, and $\ln(1) = 0$).

Species Evenness

Pielou's Evenness Index (J' or E)

- A measure of how evenly individuals are distributed among the different species in a given community
- To evaluate how evenly individuals are spread without the influence of species count, **Pielou's Evenness Index (E)** is derived from H :

$$H' = -(-1.0297) = \mathbf{1.0297}$$

$$E = \frac{H}{H_{\max}} = \frac{H}{\ln(S)}$$

Using our example ($H' = 1.0297$, $S = 3$):

1. $H_{\max} = \ln(3) \approx 1.0986$

2. $E = \frac{1.0297}{1.0986} \approx \mathbf{0.937}$

- **Value of 1:** Perfect evenness. Every single species in the community has the exact same number of individuals.
- **Value close to 0:** Extreme dominance. One species makes up almost the entire community, while the others are very rare.



THANK YOU